

SONG TREND LIFECYCLE ANALYSIS

1. PROJECT OVERVIEW

This project analyzes song streaming behavior using weekly chart data from 10,798 entries spanning 635 unique songs across 54 weeks of the Spotify India Regional Charts (May 2025 – May 2026). The goal is to classify songs by their streaming lifecycle pattern. How they rise, peak, and decline on the charts and uncover insights into artist performance, label dominance, and sustained listener engagement to offer a more complete picture of what chart success actually looks like in the Indian music market.

2. DATASET SUMMARY

- Rows: 10798
- Columns: 10
- Key Features:
 - Chart Performance (rank, peak_rank, previous_rank, weeks_on_chart)
 - Track Details (uri, artist_names, track_name, streams)
 - Source Information (source, source_file)

3. EXPLORATORY DATA ANALYSIS USING PYTHON

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	count	unique		top	freq	mean	std	min	25%	50%	75%	max
rank	10798.0	NaN		NaN	NaN	100.484442	57.730667	1.0	50.25	100.0	150.0	200.0
uri	10798	635	spotify:track:5ThyDv6aRVU8AH4vXQNlIdF	54		NaN	NaN	NaN	NaN	NaN	NaN	NaN
artist_names	10798	487	Karan Aujla, Ikky	244		NaN	NaN	NaN	NaN	NaN	NaN	NaN
track_name	10798	602	Sahiba	99		NaN	NaN	NaN	NaN	NaN	NaN	NaN
source	10798	142	T-Series	2357		NaN	NaN	NaN	NaN	NaN	NaN	NaN
peak_rank	10798.0	NaN		NaN	NaN	39.828765	43.249913	1.0	6.0	24.0	58.0	200.0
previous_rank	10798.0	NaN		NaN	NaN	87.764956	58.603985	-1.0	36.0	86.0	137.0	200.0
weeks_on_chart	10798.0	NaN		NaN	NaN	76.397203	85.836235	1.0	15.0	43.0	102.0	363.0
streams	10798.0	NaN		NaN	NaN	2243930.571217	1313574.31516	1214714.0	1525125.75	1831004.5	2426274.5	26355165.0
source_file	10798	54	regional-in-weekly-2025-05-08.csv	200		NaN	NaN	NaN	NaN	NaN	NaN	NaN

- **Missing Value:** Checked for null values and Identified No missing values in any columns.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
 - Performed deduplication on 10,798 Spotify chart entries to obtain 625 unique songs and exported the cleaned dataset to a CSV file.
 - Created **lifetime_weeks_on_chart** column by counting the number of distinct weeks each track appeared on the chart across its entire tracking period.
 - Created **last_appearance_streams** column by extracting the stream count from the final week each track appeared on the chart.
 - Created **last_appearance_rank** column by capturing the chart rank of each track during its final recorded appearance.
 - Created **last_appearance_date** column by identifying the date of the last week each track appeared on the chart.
- **Data Integration:** Connected Python script to MySQL database and loaded the cleaned DataFrame into MySQL tables for structured storage and SQL-based analysis.

4. Data Analysis using SQL

Performed structured analysis in MYSQL to answer key business questions:

1. **Chart Longevity Ranking Analysis**-Identified top songs based on total weeks spent on the chart.

	track_name	tot_weeks
►	Dil Diyan Gallan	54
	Tere Bina	54
	Die With A Smile	54
	Jeene Laga Hoon	54
	Winning Speech	54
	Ishq Hai	54
	Saiyyan	54
	Sang Rahiyo	54
	Sahiba	54
	Ranjheya Ve	54

2. Longest Chart Duration Analysis-Found songs with the longest lifetime on the chart.

	track_name	artist_names	lifetime_weeks_on_chart
▶	Agar Tum Saath Ho (From "Tamasha")	Alka Yagnik, Arijit Singh	363
	Tujhe Kitna Chahne Lage (From "Kabir Singh")	Arijit Singh, Mithoon	363
	Kaise Hua (From "Kabir Singh")	Vishal Mishra, Manoj Muntashir	361
	Pal	Javed-Mohsin, Arijit Singh, Shreya Ghoshal, Ku...	350
	Kabira	Pritam, Tochi Raina, Rekha Bhardwaj, Amitabh ...	340
	Dil Diyan Gallan	Atif Aslam	326
	Ae Dil Hai Mushkil Title Track	Pritam, Arijit Singh	326
	Shayad	Pritam, Arijit Singh	322
	Tum Se Hi	Pritam, Mohit Chauhan, Irshad Kamil	321
	Mere Sohneya (From "Kabir Singh")	Sachet Tandon, Parampara Tandon, Sachet-Par...	317

3. Fastest Peak Achievement Analysis-Analyzed songs reaching peak rank in shortest time.

track_name	artist_names	peak_rank	weeks_on_chart_in_dataset	lifetime_weeks_on_chart
▶ Tu Hain Toh Main Hoon (From "Sky Force")	Tanishk Bagchi, Arijit Singh, Afsana Khan, Irsha...	7	15	29
Tu Hain Toh (From "Mr. And Mrs. Mahi")	Bunny, Hunny, Sagar	8	15	60
Dil Tu Jaan Tu	Gurnazar, Chet Singh	14	15	55
Teri Ramjhol Bole Gi (feat. Kay D)	Masoom Sharma, Sheenam Katholic, Mr. Boota, ...	18	15	22
Deewaana Deewaana	A.R. Rahman, Irshad Kamil	19	15	15
Kithe Reh Gaya	Neeti Mohan, Abhijit Vaghani	43	15	15
By My Side	AP Dhillon, Shinda Kahlon	87	15	15
Falani	Vikram Sarkar, Guri Nimana	94	15	15
Piche Tere	Kunwar, Dishant	94	15	15
Toh Phir Aao	Pritam, Mustafa Zahid, Sayeed Quadri	155	15	15
FA9LA	Flipperachi	1	16	16
Powerhouse	Anirudh Ravichander, Arivu	13	16	18
Aadat (Feat. AP Dhillon)	Yo Yo Honey Singh, AP Dhillon	47	16	16
Tutor	Cheema Y, Gur Sidhu	49	16	16
Deewana Kar Raha Hai	Javed Ali	86	16	32
Gaadi Paache Gaadi	Amanraj Gill, Bulbul	87	16	16
I Think They Call This Love - Cover	Matthew Ifield	95	16	16
One Of The Girls (with JENNIE, Lily Rose D...)	The Weeknd, JENNIE, Lily-Rose Depp	98	16	85
Tere Sang Yaara	Atif Aslam, Manoj Muntashir	101	16	57
Aadat	Jal, Mithoon, Atif Aslam	131	16	16
Sheesha - Aakhyai Mai Aakh Ghali Jo Bairan	Mitta Ror, Swara Verma	2	17	17
Khat	Navjot Ahuja	3	17	17
Humdar (From "Ek Villain")	Arijit Singh	17	17	267
Mast Magan (From "2 States")	Arijit Singh, Chinmayi	43	17	236
Itni Si Baat Hain (From "Azhari")	Pritam, Arijit Singh, Antara Mitra, Manoj Yadav	70	17	46

track_name	artist_names	peak_rank	weeks_on_chart_in_dataset	lifetime_weeks_on_chart
▶ Itni Si Baat Hain (From "Azhari")	Pritam, Arijit Singh, Antara Mitra, Manoj Yadav	70	17	46
Woh Lamhe Woh Baatein	Roop Kumar Rathod, Mithoon, Naresh Sharma, ...	77	17	21
Udaariyan	Satinder Sartaj	79	17	17
Ilahi	Pritam, Arijit Singh, Amitabh Bhattacharya	86	17	35
Toota Jo Kabhi Tara	Sumedha Karmahe, Atif Aslam	93	17	17
Chammak Challo	Vishal-Shekhar, Akon, Hamsika Iyer, Vishal Dadl...	105	17	73
Warning	Masoom Sharma, Swara Verma	130	17	17
Sanam Teri Kasam	Himesh Reshammiya, Ankit Tiwari, Palak Muchhal	1	18	32
Pehle Bhi Main	Vishal Mishra, Raj Shekhar	1	18	92
Tera Chehra	Himesh Reshammiya, Arijit Singh	3	18	32
Minnalvala	Jakes Bejoy, Sid Sriram, Sithara Krishnakumar	8	18	20
Courtside	Karan Aujla, Signature By SB	13	18	18
Dhundhala	Yashraj, Dropped Out, Talwinder	38	18	31
Afghan Jalebi (Film Version)	Pritam, Akhtar Chinnal	114	18	18
Sanam Teri Kasam	Himesh Reshammiya, Ankit Tiwari, Palak Muchhal	1	19	51
Bekhayali (From "Kabir Singh")	Sachet Tandon, Sachet-Parampara	1	19	285
Mann Mera	Gajendra Verma	12	19	230
At Peace	Karan Aujla, Ikky	16	19	19
Aashiq Tera	Sohal Sen, Altamash Faridi	23	19	27
Jeena Jeena	Sachin-Jigar, Atif Aslam	24	19	206
Tu	Talwinder, Sanjoy	27	19	19
Tera Hone Laga Hoon	Pritam, Atif Aslam, Alisha Chinai	30	19	226
Tu Jaane Na	Pritam, Atif Aslam	48	19	206
Chikri Chikri (From "Peddi") [TELUGU]	A.R. Rahman, Mohit Chauhan, Balaji	58	19	19
Tera Mera Rishta	Pritam, Mustafa Zahid, Sayeed Quadri	88	19	27

4. Top Streaming Songs Analysis-Identified songs with highest total streams.

	track_name	total_streams
▶	Sahiba	402982566
	Finding Her	329613619
	Ishq	265319680
	Apna Bana Le	240174868
	Raanjhan (From "Do Patti")	233490985
	Jo Tum Mere Ho	230122017
	Tere Bina	214128513
	For A Reason	211845552
	Haseen	205475030
	Saiyaara	195822861

5. Rank Drop Analysis-Measured biggest decline from peak to exit rank.

	track_name	artist_names	peak_rank	last_appearance_rank	rank_drop
▶	Heeriye (feat. Arijit Singh)	Jasleen Royal, Arijit Singh, Dulquer Salmaan	1	200	199
	Satrange (From "ANIMAL")	Arijit Singh, Shreyas Puranik, Siddharth - Garima	2	199	197
	Softly	Karan Aujla, Ikky	4	200	196
	Cheques	Shubh	1	197	196
	Chuttamalle (From "Devara Part 1")	Shilpa Rao, Anirudh Ravichander, Ramajogayya...	4	200	196
	Se-Orita	Shawin Mendes, Camila Cabello	1	196	195
	O Maahi	Pritam, Arijit Singh, Irshad Kamil	4	199	195
	Tumhare Hi Rahenge Hum (From "Stree 2")	Sachin-Jigar, Varun Jain, Shilpa Rao, Amitabh B...	8	200	192
	Shaky	Sanju Rathod, G-SPXRK	3	194	191
	Akhiyaan Gulaab (From "Teri Baaton Mein Aisa ...	Mitraz	2	193	191
	Sanam Teri Kasam	Himesh Reshammiya, Ankit Tiwari, Palak Muchhal	1	192	191
	Pardesiya - From "Param Sundari"	Sachin-Jigar, Sonu Nigam, Krishnakali Saha, Ami...	4	194	190
	Janiye (From the Netflix Film "Chor Nikal Ke Bha...	Vishal Mishra, Rashmeet Kaur	5	195	190
	Jaana Samjho Na (From "Bhool Bhulaiyaa 3")	Aditya Rikhari, Tulsi Kumar, Lijo George-DJ Chetas	9	199	190
	Dekhha Tenu (From "Mr. And Mrs. Mahi")	Mohammad Faiz, Jaani	2	190	188
	Ranjha (From "Sherashaah")	Jasleen Royal, B Praak, Romy	1	187	186
	Zaroor	Aparshakti Khurana, Savi Kahlon	13	198	185
	Excuses	AP Dhillon, Gurinder Gill, Intense	1	186	185
	Aavan Jaavan	Pritam, Arijit Singh, Nikhita Gandhi, Amitabh Bha...	8	192	184
	Sirra	Guru Randhawa, Kiran Bajwa, Rony Ajnali	6	189	183
	Supreme	Shubh	2	185	183
	Laal Pari (From "Housefull 5")	Yo Yo Honey Singh, Simar Kaur, Alfaaz	9	192	183
	Suniyan Suniyan	Juss, MxSingh	3	186	183
	I-Don't Know - I-Popstar: Vol. 1	Maan Panu	6	189	183
	Ve Kamleya (From "Rocky Aur Rani Ki Prem Kah...	Pritam, Arijit Singh, Shreya Ghoshal, Shadab Fa...	15	198	183

	track_name	artist_names	peak_rank	last_appearance_rank	rank_drop
	Aasa Kooda - From "Think Indie"	Sai Abhyankkar, Sai Smriti	3	185	182
	Powerhouse	Anirudh Ravichander, Arivu	13	194	181
	Aayi Nai (From "Stree 2")	Sachin-Jigar, Pawan Singh, Simran Choudhary, ...	2	181	179
	2 Khatole (Lo-fi Mix)	Masoom Sharma	17	196	179
	Wavy	Karan Aujla, Jay Trak	4	182	178
	AZUL	Guru Randhawa, Gurjit Gill, Lavish Dhiman	10	188	178
	Qatal	Guru Randhawa, Sanjoy, Gill Madhrai	4	182	178
	52 Bars	Karan Aujla, Ikky	12	190	178
	MF Gabhrui	Karan Aujla, Ikky	21	199	178
	Saiyaara Reprise - Female	Tanishk Bagchi, Faheem Abdullah, Arslan Nizami...	5	183	178
	STFU	AP Dhillon, Shinda Kahlon	15	192	177
	Monica	Anirudh Ravichander, Sublahshini, Asal Kolaar	12	189	177
	Minnalvala	Jakes Bejoy, Sid Sriram, Sithara Krishnakumar	8	184	176
	Tum Se (From "Teri Baaton Mein Aisa Uljha Jiya")	Sachin-Jigar, Raghav Chaitanya, Varun Jain, In...	10	185	175
	Courtside	Karan Aujla, Signature By SB	13	187	174
	Aaj Ki Raat (From "Stree 2")	Sachin-Jigar, Madhubanti Bagchi, Divya Kumar, ...	1	174	173
	Desi Kalakaar	Yo Yo Honey Singh	13	186	173
	Chambal K Dakku	Masoom Sharma, Rahul Muana, Swara Verma, ...	26	198	172
	Ajab Si	KK, Vishal-Shekhar, Vishal Dadlani	19	191	172
	Dil Tu Jaan Tu	Gurnazar, Chet Singh	14	186	172
	Perfect	Ed Sheeran	29	200	171
	Pehle Bhi Main	Vishal Mishra, Raj Shekhar	1	172	171
	Winning Speech	Karan Aujla, Mxrd	8	177	169
	O Saathi	Atif Aslam, Arko	28	197	169
	Shayad	Pritam, Arijit Singh	1	170	169

6. Rank Improvement Analysis-Found songs with highest week-over-week improvement.

	track_name	artist_names	previous_rank	peak_rank	rank_improvement
►	Fell For You	Shubh	190	19	171
	Kalank - Title Track	Pritam, Arijit Singh	167	5	162
	Dil Tu Jaan Tu	Gurnazar, Chet Singh	174	14	160
	Desi Kalakaar	Yo Yo Honey Singh	173	13	160
	Sanam Teri Kasam	Himesh Reshammiya, Ankit Tiwari, Palak Muchhal	156	1	155
	Tu Hain Toh (From "Mr. And Mrs. Mahi")	Bunny, Hunny, Sagar	162	8	154
	Tutor	Cheema Y, Gur Sidhu	197	49	148
	Dil Nu	AP Dhillon, Shinda Kahlon	181	33	148
	Dil Ibaadat	Pritam, KK, Sayeed Quadri	197	54	143
	Starboy	The Weeknd, Daft Punk	150	12	138

7. Artist Popularity Analysis (Song Count)-Analyzed artists with most charting songs.

	artist_names	total_charting_songs
►	Karan Aujla, Ikky	7
	Pritam, Arijit Singh	5
	Shubh	4
	Anuv Jain	4
	Gajendra Verma	3
	Aditya Rikhari	3
	Pritam, KK, Sayeed Quadri	3
	AP Dhillon, Shinda Kahlon	3
	Yo Yo Honey Singh	3
	Dhanda Nyoliwala	2
	Pritam, Atif Aslam, Alisha ...	2
	Kailash Kher, Paresh Kam...	2
	Himesh Reshammiya, Arij...	2
	Arijit Singh, Mithoon	2
	Pritam, Mohit Chauhan, Ir...	2
	Sachin-Jigar, Atif Aslam	2
	Arijit Singh, Chinmayi	2
	Arijit Singh	2
	Pritam, Atif Aslam	2
	Sohail Sen, Altamash Faridi	2

8. Early Peak vs Weak Exit Analysis-Found songs with strong start but weak ending.

	track_name	artist_names	peak_rank	last_appearance_rank	rank_drop
▶	Heeriye (feat. Arijit Singh)	Jasleen Royal, Arijit Singh, Dulquer Salmaan	1	200	199
	Satranga (From "ANIMAL")	Arijit Singh, Shreyas Puranik, Siddharth - Garima	2	199	197
	Softly	Karan Aujla, Ikky	4	200	196
	Cheques	Shubh	1	197	196
	Chuttamalle (From "Devara Part 1")	Shilpa Rao, Anirudh Ravichander, Ramajogayya...	4	200	196
	O Maahi	Pritam, Arijit Singh, Irshad Kamil	4	199	195
	Se-ori	Shawn Mendes, Camila Cabello	1	196	195
	Tumhare Hi Rahenge Hum (From "Stree 2")	Sachin-Jigar, Varun Jain, Shilpa Rao, Amitabh B...	8	200	192
	Shaky	Sanju Rathod, G-SPXRK	3	194	191
	Sanam Teri Kasam	Himesh Reshammiya, Ankit Tiwari, Palak Muchhal	1	192	191
	Akhiyaan Gulaab (From "Teri Baaton Mein Ai...	Mitraz	2	193	191
	Janiye (from the Netflix Film "Chor Nikal Ke ...	Vishal Mishra, Rashmeet Kaur	5	195	190
	Pardesiya - From "Param Sundari"	Sachin-Jigar, Sonu Nigam, Krishnakali Saha, Ami...	4	194	190
	Jaana Samjho Na (From "Bhool Bhulaiyaa 3")	Aditya Rikhi, Tulsi Kumar, Lijo George-Dj Chetas	9	199	190
	Dekhha Tenu (From "Mr. And Mrs. Mahi")	Mohammad Faiz, Jaani	2	190	188
	Ranjha (From "Shershaah")	Jasleen Royal, B Praak, Romy	1	187	186
	Excuses	AP Dhillon, Gurinder Gill, Intense	1	186	185
	Zaroor	Aparshakti Khurana, Savi Kahlon	13	198	185
	Aavan Jaavan	Pritam, Arijit Singh, Nikhita Gandhi, Amitabh Bha...	8	192	184
	Laal Pari (From "Housefull 5")	Yo Yo Honey Singh, Simar Kaur, Alfaaz	9	192	183

9. Stability Analysis-Identified songs with minimal rank fluctuation.

Result Grid						
Filter Rows:		Exports	Wrap Cell Contents	Fetch rows:		
	track_name	artist_names	peak_rank	last_appearance_rank	lifetime_weeks_on_chart	rank_volatility
▶	Deewana Deewana	A.R. Rahman, Irshad Kamil	19	19	15	0
	Khat	Nayot Ahuja	3	3	17	0
	Chahun Main Ya Naa (From "Aashiqui 2")	Palak Muchhal, Arijit Singh	41	41	134	0
	Itni Si Baat Hain (From "Aashiqui 2")	Pritam, Arijit Singh, Antara Mitra, Manoj Yadav	70	70	46	0
	Darkhaast	Mithoon, Arijit Singh, Sunidhi Chauhan, Sayeed ...	18	18	24	0
	Gehra Hua	Shashwat Sachdev, Arijit Singh, Irshad Kamil, A...	1	2	24	1
	Phir Mohabbat	Mohammed Irfan, Arijit Singh, Sam Bhat, Mitho...	85	86	69	1
	Arz Kiya Hai Coke Studio Bharat	Anuv Jain	5	7	38	2
	Mann Mera	Gajendra Verma	12	14	230	2
	Aaj Se Teri	Amit Trivedi, Arijit Singh	114	117	19	3
	Hale Dil	Harshit Saxena, Sayeed Quadri	60	63	80	3
	Sheesha - Aakhiyaan Gulaab	Mitraz	2	6	17	4
	Sun Saathiya	Divya Kumar, Priya Saraiya	110	114	51	4
	Teri Deewani	Kailash Kher, Pooja Kamath, Nareesh Kamath	28	33	93	5
	Sitaare (From "Ekkees")	Arijit Singh, Sachin-Jigar, Amitabh Bhattacharya	8	13	23	5
	Mann Mera	Gajendra Verma	7	14	265	7
	Samjhawan	Jawad Ahmad, Shaarib Toshi, Arijit Singh, Shre...	3	11	247	8
	Finding Her	Kushagra, Bharath, Saheal	1	9	62	8
	Humsafar (From "Badrinath Ki Dulhania")	Alkhil Sachdeva, Mansheel Gujral	44	53	121	9
	Sahiba	Aditya Rikhi	1	10	45	9

10. Song Lifecycle Classification-Categorized songs into lifecycle patterns (Flash Hit, etc.).

	track_name	artist_names	peak_rank	last_appearance_rank	weeks_on_chart_in_dataset	lifetime_weeks_on_chart	total_streams	lifecycle_pattern
▶	Dil Diyan Gallan	Atif Aslam	29	134	54	326	114820648	Slow Burn
	Tere Bina	A.R. Rahman, Chinmayi, Murtuza Khan, Qadir K...	8	31	54	91	214128513	Standard
	Die With A Smile	Lady Gaga, Bruno Mars	4	145	54	85	99941243	Standard
	Jeene Laga Hoon	Sachin-Jigar, Atif Aslam, Shreya Ghoshal	28	141	54	88	113741416	Slow Burn
	Winning Speech	Karan Aujla, Mxci	8	177	54	104	114807524	Standard
	Ishq Hai	Anurag Saikia, Raj Shekhar, Romy, Amarabha B...	1	149	54	74	115720616	Standard
	Saiyyan	Kailash Kher, Pooja Kamath, Nareesh Kamath	54	169	54	124	92181564	Slow Burn
	Sang Rahiyo	Jasleen Royal, Ujjwal Kashyap, Neeraj Rajawat	19	77	54	114	134518042	Slow Burn
	Sahiba	Jasleen Royal, Stebin Ben, Vijay Deverakonda, ...	1	123	54	77	131848424	Standard
	Ranjheya Ve	Zain Zohab	20	81	54	64	123798934	Slow Burn
	Husn	Anuv Jain	1	124	54	128	11727211	Standard
	Baarishein	Anuv Jain	16	138	54	261	102957931	Slow Burn
	Softly	Karan Aujla, Ikky	4	200	54	143	103533935	Standard
	Tum Se Hi	Pritam, Mohit Chauhan, Irshad Kamil	12	64	54	321	113008931	Slow Burn
	Starboy	The Weeknd, Daft Punk	12	122	54	250	83615569	Slow Burn
	Ishq Di Baajiyaan	Shankar-Ehsaan-Loy, Diljit Dosanjh	60	137	54	90	102167756	Slow Burn
	Aopia Bana Le	Sachin-Jigar, Arijit Singh, Amitabh Bhattacharya	1	12	54	151	240174868	Sustained Peak
	Ye Tune Kya Kiya	Pritam, Javed Bashir, Rajat Arora	9	34	54	117	179504679	Standard
	Pehli Dafa	Atif Aslam, Shiraz Uppal, Shakeel Sohail	40	85	54	132	115011281	Slow Burn
	Hasi - Male Version	Ami Mishra, Kunaal Vermaa	59	119	54	150	97825014	Slow Burn
	Teri Deewani	Kailash Kher, Pooja Kamath, Nareesh Kamath	28	33	54	93	135861071	Slow Burn
	Finding Her	Kushagra, Bharath, Saheal	1	9	54	62	329613619	Sustained Peak
	Russian Bandana	Dhanda Nyoliwala	6	147	54	85	99592118	Standard
	Raanjhanaa (Fr...	A.R. Rahman, Jaswinder Singh, Shiraz Uppal	50	65	54	91	109580593	Slow Burn
	Zaalima	Arijit Singh, Harshdeep Kaur	27	37	54	151	132775732	Slow Burn
	...	...	...	...	...	...	...	...

11. Long-Lived Source Analysis-Analyzed sources with most long-lasting songs.

	source	total_songs	long_lived_songs	long_lived_pct	avg_lifetime_weeks
▶	T-Series	59	59	100.0	127.2
	Sony Music Entertainment India Pvt. Ltd.	33	33	100.0	127.8
	Zee Music Company	16	16	100.0	89.6
	Saregama India Ltd	15	15	100.0	44.5
	Universal Music India Pvt Ltd.	11	11	100.0	42.5
	YRF Music	10	10	100.0	85.2
	Tips Music Limited	9	9	100.0	147.3
	Warner Music India	7	7	100.0	63.6
	WM Canada	7	7	100.0	61.1
	Sony Music Entertainment India Pvt. Ltd. / ERO...	6	6	100.0	105.2
	Sony Music / EROS Music	5	5	100.0	126.0
	SHUBH	4	4	100.0	96.3
	Sony Music India / Nine	4	4	100.0	43.8
	Sony Music / Vishesh Films	4	4	100.0	19.8
	Universal Music India Pvt. Ltd. (on behalf of An...	3	3	100.0	160.3
	All Good Music Pvt Ltd	3	3	100.0	30.3
	AP Dhillon	3	3	100.0	29.3
	Think Music	3	3	100.0	45.0

12. Artist Volume Analysis-Found artists with highest number of songs.

	artist_names	total_songs
▶	Karan Aujla, Ikky	7
	Pritam, Arijit Singh	5
	Shubh	4
	Anuv Jain	4
	Gajendra Verma	3
	Aditya Rikhari	3
	Pritam, KK, Sayeed Quadri	3
	AP Dhillon, Shinda Kahlon	3
	Yo Yo Honey Singh	3
	Dhanda Nyoliwala	2
	Pritam, Atif Aslam, Alisha ...	2
	Kailash Kher, Paresh Kam...	2
	Himesh Reshammiya, Arijit...	2
	Arijit Singh, Mithoon	2
	Pritam, Mohit Chauhan, Ir...	2
	Sachin-Jigar, Atif Aslam	2
	Arijit Singh, Chinmayi	2
	Arijit Singh	2
	Pritam, Atif Aslam	2
	Sohail Sen, Altamash Faridi	2

13. Number One Hits Analysis-Identified songs that reached rank #1.

	track_name	artist_names	peak_week_streams	lifetime_weeks_on_chart
▶	Sahiba	Aditya Rikhari	12858656	45
	Saiyaara	Tanishk Bagchi, Faheem Abdullah, Arslan Nizami...	10765955	48
	FA9LA	Flipperachi	10405970	16
	Gehra Hua	Shashwat Sachdev, Arijit Singh, Irshad Kamil, A...	9854486	24
	Finding Her	Kushagra, Bharath, Saaheal	8770636	62
	Raanjhan (From "Do Patti")	Sachet-Parampara, Parampara Tandon, Kausar ...	6023059	82
	Apna Bana Le	Sachin-Jigar, Arijit Singh, Amitabh Bhattacharya	5682005	151
	Jo Tum Mere Ho	Anuv Jain	5179390	92
	Ishq Hai	Anurag Saikia, Raj Shekhar, Romy, Amarabha B...	4140901	74
	Sahiba	Jasleen Royal, Stebin Ben, Vijay Deverakonda, ...	3469840	77
	Sajni (From "Laapataa Ladie...)	Ram Sampath, Arijit Singh, Prashant Pandey	3243608	106
	Husn	Anuv Jain	2990116	128
	Cheques	Shubh	2665646	148
	Ranjha (From "Shershaah")	Jasleen Royal, B Praak, Romy	2570937	239
	Aaj Ki Raat (From "Street 2")	Sachin-Jigar, Madhubanti Bagchi, Divya Kumar, ...	2568902	85
	Sanam Teri Kasam	Himesh Reshammiya, Ankit Tiwari, Palak Muchhal	2166219	32
	Shayad	Pritam, Arijit Singh	2146419	322
	Pehle Bhi Main	Vishal Mishra, Raj Shekhar	2099714	92
	Kesariya	Pritam, Arijit Singh, Amitabh Bhattacharya	1954914	177
	Heeriye (feat. Arijit Singh)	Jasleen Royal, Arijit Singh, Dulquer Salmaan	1881963	112
	Se-ori	Shawn Mendes, Camila Cabello	1831731	239
	Excuses	AP Dhillon, Gurinder Gill, Intense	1653046	210
	Sanam Teri Kasam	Himesh Reshammiya, Ankit Tiwari, Palak Muchhal	1571443	51
	Bekhayali (From "Kabir Sing...)	Sachet Tandon, Sachet-Parampara	1553312	285

14. Source-wise Lifetime Analysis-Calculated average song lifespan by source.

	source	total_songs	avg_lifetime_weeks	avg_weeks_in_dataset	longest_chart_run
▶	Atlantic Records UK	2	286.0	33.5	300
	The Weeknd/Lyric	1	250.0	54.0	250
	Domino Recording Co	1	177.0	48.0	177
	Universal Music India Pvt. Ltd. (on behalf of An...	3	160.3	54.0	261
	CollabX Productions	1	160.0	54.0	160
	RUN-UP RECORDS	2	158.0	32.0	210
	Island Records	2	154.5	26.5	239
	Tips Music Limited	9	147.3	34.1	261
	Sony Music Entertainment India Pvt. Ltd.	33	127.8	42.2	350
	T-Series	59	127.2	34.4	363
	Sony Music / EROS Music	5	126.0	30.8	265
	Sony Music Entertainment India Pvt. Ltd. / ERO...	6	105.2	23.8	243
	Karan Aujla	1	104.0	54.0	104
	Netflix Music	2	100.5	43.0	127
	SHUBH	4	96.3	44.8	148
	Zee Music Company	16	89.6	30.8	301
	YRF Music	10	85.2	41.5	326
	Interscope Records	1	85.0	54.0	85
	XO / Republic Records	1	85.0	16.0	85
	Artiste First	2	80.5	52.0	111
	Mix It Up Studio	1	80.0	30.0	80
	Speed Records	1	78.0	49.0	78
	Universal Music India Pvt. Ltd. (on behalf of An...	1	67.0	54.0	67
	Zain Zohaib	1	64.0	54.0	64

15. Rank Jump Analysis-Found songs with highest rank improvement.

	track_name	artist_names	previous_rank	peak_rank	rank_jump
▶	Fell For You	Shubh	190	19	171
	Kalank - Title Track	Pritam, Arijit Singh	167	5	162
	Desi Kalakaar	Yo Yo Honey Singh	173	13	160
	Dil Tu Jaan Tu	Gurnazar, Chet Singh	174	14	160
	Sanam Teri Kasam	Himesh Reshammiya, Ankit Tiwari, Palak Muchhal	156	1	155
	Tu Hain Toh (From "Mr. And Mrs. Mahi")	Bunny, Hunny, Sagar	162	8	154
	Dil Nu	AP Dhillon, Shinda Kahlon	181	33	148
	Tutor	Cheema Y, Gur Sidhu	197	49	148
	Dil Ibaadat	Pritam, KK, Sayeed Quadri	197	54	143
	Starboy	The Weeknd, Daft Punk	150	12	138
	Excuses	AP Dhillon, Gurinder Gill, Intense	137	1	136
	Zaroor	Aparshakti Khurana, Savi Kahlon	141	13	128
	52 Bars	Karan Aujla, Ikky	136	12	124
	Aasa Kooda - From "Think Indie"	Sai Abhyankar, Sai Smriti	124	3	121
	Shayad	Pritam, Arijit Singh	112	1	111
	Perfect	Ed Sheeran	139	29	110
	Aashiq Tera	Sohail Sen, Altamash Faridi	133	23	110
	Tera Chehra	Himesh Reshammiya, Arijit Singh	113	3	110
	Admirin' You (feat. Preston Pablo)	Karan Aujla, Ikky, Preston Pablo	131	24	107
	Tere Bina Na Guzara E	Josh Brar	119	19	100

16. Longevity Distribution Analysis-Analyzed songs with 50+ and 100+ weeks presence.

track_name	artist_names	source	lifetime_weeks_on_chart	longevity_tier
Agar Tum Saath Ho (From "Tamasha")	Alka Yagnik, Arijit Singh	T-Series	363	100+ weeks
Tujhe Kitna Chahne Lage (From "Kabir Singh")	Arijit Singh, Mithoon	T-Series	363	100+ weeks
Kaise Hua (From "Kabir Singh")	Vishal Mishra, Manoj Muntashir	T-Series	361	100+ weeks
Pal	Javed-Mohsin, Arijit Singh, Shreya Ghoshal, Ku...	Sony Music Entertainment India Pvt. Ltd.	350	100+ weeks
Kabira	Pritam, Tochi Raina, Rekha Bhardwaj, Amitabh ...	T-Series	340	100+ weeks
Dil Diyan Gallan	Atif Aslam	YRF Music	326	100+ weeks
Ae Dil Hai Mushkil Title Track	Pritam, Arijit Singh	Sony Music Entertainment India Pvt. Ltd.	326	100+ weeks
Shayad	Pritam, Arijit Singh	Sony Music Entertainment India Pvt. Ltd.	322	100+ weeks
Tum Se Hi	Pritam, Mohit Chauhan, Irshad Kamil	T-Series	321	100+ weeks
Mere Sohneya (From "Kabir Singh")	Sachet Tandon, Parampara Tandon, Sachet-Par...	T-Series	317	100+ weeks
Humdard (From "Ek Villain")	Arijit Singh	T-Series	304	100+ weeks
Kalank - Title Track	Pritam, Arijit Singh	Zee Music Company	301	100+ weeks
Perfect	Ed Sheeran	Atlantic Records UK	300	100+ weeks
Bekhayali (From "Kabir Singh")	Sachet Tandon, Sachet-Parampara	T-Series	285	100+ weeks
Saibo	Sachin-Jigar, Shreya Ghoshal, Tochi Raina	Sony Music Entertainment India Pvt. Ltd.	277	100+ weeks
Mast Magan (From "2 States")	Arijit Singh, Chinmayi	T-Series	273	100+ weeks
Shape of You	Ed Sheeran	Atlantic Records UK	272	100+ weeks
Humdard (From "Ek Villain")	Arijit Singh	T-Series	267	100+ weeks
Mann Mera	Gajendra Verma	Sony Music / EROS Music	265	100+ weeks
Tum Hi Ho	Arijit Singh, Mithoon	T-Series	264	100+ weeks
Baarishein	Anuv Jain	Universal Music India Pvt. Ltd. (on behal...	261	100+ weeks

17. Overall Chart Duration Analysis-Measured average and range of song lifetimes.

Result Grid					
	avg_weeks_in_dataset	avg_lifetime_weeks	min_weeks	max_weeks	total_songs
▶	34.7	93.1	15	54	258

18. Best Performing Artists Analysis-Identified artists with best average peak rank.

artist_names	total_songs	avg_peak_rank	best_peak	number_one_hits
▶ Himesh Reshammiya, Ankit Tiwari, Palak Muchhal	2	1.0	1	2
Himesh Reshammiya, Arijit Singh	2	3.0	3	0
Anuv Jain	4	5.8	1	2
Shubh	4	6.3	1	1
Gajendra Verma	3	13.0	7	0
Aditya Rikhari	3	14.0	1	1
Arijit Singh, Mithoon	2	14.0	2	0
Karan Aujla, Ikky	7	15.3	2	0
Arijit Singh	2	16.0	15	0
Yo Yo Honey Singh	3	16.0	3	0
Sohail Sen, Altamash Faridi	2	23.0	23	0
Sachin-Jigar, Atif Aslam	2	24.0	24	0
Pritam, Mohit Chauhan, Irshad Kamil	2	24.5	12	0
Pritam, Atif Aslam, Alisha Chinai	2	30.0	30	0
Pritam, Arijit Singh	5	31.2	1	1
Arijit Singh, Chinmayi	2	35.5	28	0
Cheema Y, Gur Sidhu	2	38.0	27	0
Dhanda Nyoliwala	2	40.0	6	0
Kailash Kher, Paresh Kamath, Naresh Kamath	2	41.0	28	0
Pritam, Arijit Singh, Amitabh Bhattacharya	2	43.5	1	1

19. Repeat Appearance Analysis-Found songs appearing multiple times in dataset.

	track_name	artist_names	total_weeks_in_dataset
►	Humdard (From "Ek Villain")	Arijit Singh	54
	Aashiq Tera	Sohail Sen, Altamash Faridi	54
	Tera Hone Laga Hoon	Pritam, Atif Aslam, Alisha Chinai	54
	Mann Mera	Gajendra Verma	54
	Tu Jaane Na	Pritam, Atif Aslam	54
	Tera Chehra	Himesh Reshammiya, Arijit Singh	53
	Jeena Jeena	Sachin-Jigar, Atif Aslam	53
	Sanam Teri Kasam	Himesh Reshammiya, Ankit Tiwari, Palak Muchhal	37

20. Platform Distribution Analysis-Analyzed song distribution across sources.

	source	total_songs	pct_of_total
►	T-Series	59	22.9
	Sony Music Entertainment India Pvt. Ltd.	33	12.8
	Zee Music Company	16	6.2
	Saregama India Ltd	15	5.8
	Universal Music India Pvt Ltd.	11	4.3
	YRF Music	10	3.9
	Tips Music Limited	9	3.5
	Warner Music India	7	2.7
	WM Canada	7	2.7
	Sony Music Entertainment India Pvt. Ltd. / ERO...	6	2.3
	Sony Music / EROS Music	5	1.9
	Sony Music India / Nine	4	1.6
	SHUBH	4	1.6
	Sony Music / Vishesh Films	4	1.6
	All Good Music Pvt Ltd	3	1.2
	AP Dhillon	3	1.2
	Universal Music India Pvt. Ltd. (on behalf of An...	3	1.2
	Think Music	3	1.2
	Netflix Music	2	0.8
	EYP Creations	2	0.8

21. Top Streaming Performance Analysis-Identified songs with maximum streams.

	track_name	streams
►	Sahiba	402982566
	Finding Her	329613619
	Ishq	265319680
	Apna Bana Le	240174868
	Raanjhan (From "Do Patti")	233490985
	Jo Tum Mere Ho	230122017
	Tere Bina	214128513
	For A Reason	211845552
	Haseen	205475030
	Saiyaara	195822861

22. Number One Rank Analysis-Found songs that hit #1 position.

	track_name	peak_rank
►	Ishq Hai	1
	Sahiba	1
	Husn	1
	Apna Bana Le	1
	Finding Her	1
	Sajni (From "Laapataa Ladies")	1
	Jo Tum Mere Ho	1
	Ranjhan (From "Do Patti")	1
	Cheques	1
	Shayad	1
	Aaj Ki Raat (From "Stree 2")	1
	Ranjha (From "Shershaah")	1
	Saiyaara	1
	Kesariya	1
	Señorita	1
	Excuses	1
	Gehra Hua	1
	Heeriye (feat. Arijit Singh)	1
	Sanam Teri Kasam	1
	Bekhayali (From "Kabir Singh")	1

23. Longest Chart Presence Analysis-Analyzed songs with highest lifetime weeks.

	track_name	lifetime_weeks_on_chart
►	Agar Tum Saath Ho (From "Tamasha")	363
	Tujhe Kitna Chahne Lage (From "Kabir Singh")	363
	Kaise Hua (From "Kabir Singh")	361
	Pal	350
	Kabira	340
	Dil Diyan Gallan	326
	Ae Dil Hai Mushkil Title Track	326
	Shayad	322
	Tum Se Hi	321
	Mere Sohneya (From "Kabir Singh")	317
	Humdard (From "Ek Villain")	304
	Kalank - Title Track	301
	Perfect	300
	Bekhayali (From "Kabir Singh")	285
	Saibo	277
	Mast Magan (From "2 States")	273
	Shape of You	272
	Humdard (From "Ek Villain")	267
	Mann Mera	265
	Tum Hi Ho	264

24. Fastest Peak Analysis-Identified songs reaching peak fastest.

	track_name	peak_rank	weeks_on_chart_in_dataset
►	Tu Hain Toh Main Hoon (From "Sky Force")	7	15
	Tu Hain Toh (From "Mr. And Mrs. Mahi")	8	15
	Dil Tu Jaan Tu	14	15
	Teri Ramjhori Bole Gi (feat. Kay D)	18	15
	Deewana Deewana	19	15
	Kithe Reh Gaya	43	15
	By My Side	87	15
	Falani	94	15
	Piche Tere	94	15
	Toh Phir Aao	155	15
	FA9LA	1	16
	Powerhouse	13	16
	Aadat (Feat. AP Dhillon)	47	16
	Tutor	49	16
	Deewana Kar Raha Hai	86	16
	Gaadi Paache Gaadi	87	16
	I Think They Call This Love - Cover	95	16
	One Of The Girls (with JENNIE, Lily Rose D...)	98	16
	Tere Sang Yaara	101	16

25. Seasonal Debut Trend Analysis-Analyzed monthly new song release patterns.

	debut_month	month_number	new_debuts
►	January	1	19
	February	2	21
	March	3	15
	April	4	15
	May	5	22
	June	6	17
	July	7	29
	August	8	31
	September	9	11
	October	10	24
	November	11	25
	December	12	29